

```
In [1]: pip install python-docx nltk pandas
```

```
Requirement already satisfied: python-docx in c:\users\first part 2026\anaconda3\lib\site-packages (1.2.0)
Requirement already satisfied: nltk in c:\users\first part 2026\anaconda3\lib\site-packages (3.9.2)
Requirement already satisfied: pandas in c:\users\first part 2026\anaconda3\lib\site-packages (2.3.3)
Requirement already satisfied: lxml>=3.1.0 in c:\users\first part 2026\anaconda3\lib\site-packages (from python-docx) (5.3.0)
Requirement already satisfied: typing_extensions>=4.9.0 in c:\users\first part 2026\anaconda3\lib\site-packages (from python-docx) (4.15.0)
Requirement already satisfied: click in c:\users\first part 2026\anaconda3\lib\site-packages (from nltk) (8.2.1)
Requirement already satisfied: joblib in c:\users\first part 2026\anaconda3\lib\site-packages (from nltk) (1.5.2)
Requirement already satisfied: regex>=2021.8.3 in c:\users\first part 2026\anaconda3\lib\site-packages (from nltk) (2025.9.1)
Requirement already satisfied: tqdm in c:\users\first part 2026\anaconda3\lib\site-packages (from nltk) (4.67.1)
Requirement already satisfied: numpy>=1.26.0 in c:\users\first part 2026\anaconda3\lib\site-packages (from pandas) (2.3.5)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\first part 2026\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\first part 2026\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\first part 2026\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\first part 2026\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: colorama in c:\users\first part 2026\anaconda3\lib\site-packages (from click->nltk) (0.4.6)
Note: you may need to restart the kernel to use updated packages.
```

```
In [2]: import nltk
import pandas as pd
from docx import Document
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
from collections import Counter
```

```
In [3]: nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
```

```

[nltk_data] Downloading package punkt to C:\Users\First Part
[nltk_data] 2026\AppData\Roaming\nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to C:\Users\First Part
[nltk_data] 2026\AppData\Roaming\nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package stopwords to C:\Users\First Part
[nltk_data] 2026\AppData\Roaming\nltk_data...
[nltk_data] Package stopwords is already up-to-date!

```

Out[3]: True

In [4]: `file_path = "big_data_technical_content_3_pages.docx"`

```

doc = Document(file_path)

text = ""

for para in doc.paragraphs:
    text += para.text + " "

print("File Loaded Successfully\n")
print(text[:500])

```

File Loaded Successfully

Technical Content for NLP Experiment (Approx. 3 Pages) Big Data Technologies and Modern Data Processing Big Data refers to extremely large datasets that cannot be efficiently processed using traditional data processing tools and database systems. With the rapid growth of digital technologies, organizations generate massive volumes of data every second. This data comes from multiple sources such as social media platforms, online transactions, mobile applications, sensors, and Internet of Things (

In [5]: `tokens = word_tokenize(text)`

```

print("Total Tokens:", len(tokens))

print("\nFirst 50 Tokens:\n")
print(tokens[:50])

```

Total Tokens: 956

First 50 Tokens:

```

['Technical', 'Content', 'for', 'NLP', 'Experiment', '(', 'Approx', '.', '3', 'Page', 's', ')', 'Big', 'Data', 'Technologies', 'and', 'Modern', 'Data', 'Processing', 'Big', 'Data', 'refers', 'to', 'extremely', 'large', 'datasets', 'that', 'can', 'not', 'be', 'efficiently', 'processed', 'using', 'traditional', 'data', 'processing', 'tools', 'and', 'database', 'systems', '.', 'With', 'the', 'rapid', 'growth', 'of', 'digital', 'technologies', ',', 'organizations', 'generate']

```

In [6]: `stop_words = set(stopwords.words('english'))`

```

filtered_tokens = [word.lower() for word in tokens if word.lower() not in stop_words]

print("Total Words After Stopword Removal:", len(filtered_tokens))

```

```
print("\nFirst 50 Words After Stopword Removal:\n")
print(filtered_tokens[:50])
```

Total Words After Stopword Removal: 598

First 50 Words After Stopword Removal:

```
['technical', 'content', 'nlp', 'experiment', 'approx', 'pages', 'big', 'data', 'technologies', 'modern', 'data', 'processing', 'big', 'data', 'refers', 'extremely', 'large', 'datasets', 'efficiently', 'processed', 'using', 'traditional', 'data', 'processing', 'tools', 'database', 'systems', 'rapid', 'growth', 'digital', 'technologies', 'organizations', 'generate', 'massive', 'volumes', 'data', 'every', 'second', 'data', 'comes', 'multiple', 'sources', 'social', 'media', 'platforms', 'online', 'transactions', 'mobile', 'applications', 'sensors']
```

```
In [7]: stemmer = PorterStemmer()

sample_words = filtered_tokens[:10]

stemmed_words = []

for word in sample_words:
    stemmed_words.append(stemmer.stem(word))

print("Original Words:\n", sample_words)

print("\nStemmed Words:\n", stemmed_words)
```

Original Words:

```
['technical', 'content', 'nlp', 'experiment', 'approx', 'pages', 'big', 'data', 'technologies', 'modern']
```

Stemmed Words:

```
['technic', 'content', 'nlp', 'experi', 'approx', 'page', 'big', 'data', 'technolog', 'modern']
```

```
In [8]: technical_terms = ["data", "hadoop", "spark", "kafka", "processing"]

term_frequency = {}

for term in technical_terms:
    term_frequency[term] = filtered_tokens.count(term)

print("Technical Term Frequency:\n")
print(term_frequency)
```

Technical Term Frequency:

```
{'data': 49, 'hadoop': 8, 'spark': 6, 'kafka': 5, 'processing': 15}
```

```
In [9]: ranked_terms = sorted(term_frequency.items(), key=lambda x: x[1], reverse=True)

print("Ranking of Technical Terms:\n")

for rank, (term, freq) in enumerate(ranked_terms, start=1):
    print(rank, ".", term, "=", freq)
```

Ranking of Technical Terms:

```
1 . data = 49
2 . processing = 15
3 . hadoop = 8
4 . spark = 6
5 . kafka = 5
```

```
In [10]: df = pd.DataFrame(ranked_terms, columns=["Technical Term","Frequency"])

df["Rank"] = df["Frequency"].rank(ascending=False, method="dense").astype(int)

print("\nTechnical Term Ranking Table\n")

print(df)
```

Technical Term Ranking Table

	Technical Term	Frequency	Rank
0	data	49	1
1	processing	15	2
2	hadoop	8	3
3	spark	6	4
4	kafka	5	5

```
In [ ]:
```